

HW1 — MEMM POS Tagging — Report

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1. Optimization & Parameter Selection

I trained a trigram MEMM via L-BFGS. The baseline global threshold was too blunt for strict caps, discarding rare-but-informative features. I implemented a per-family threshold dict to prune independently and steer the budget. Sweeping these thresholds maximized accuracy under the limits: Model 1 efficiently reached $9,814 \leq 10,000$ parameters, while Model 2 was hyper-optimized to $486 \leq 500$.

2. Feature Engineering

Beyond the baseline (word, τ), I engineered the following high-impact families:

- Lexical: Suffixes and prefixes up to length 4 (captures morphology and OOV).
- Context: Trigram/bigram/unigram tag histories, plus adjacent words.
- Orthography: Custom flags for uppercase, digits, acronyms, proper nouns, and hyphens.
- Word Shape: Maps characters to X/x/d (e.g., Apple $\rightarrow$ Xx). Crucial for OOV numbers/names.
- Word Cluster (M2): Unsupervised class ID via neighbor co-occurrence $\rightarrow$ SVD $\rightarrow$ k-means. Generalizes to unseen biomedical terms without the performance penalty of train1's domain shift.
- Back-off: Lower-cased word to share evidence.

3. Configurations & Inference

- Model 1 ($\lambda=0.3$): Utilizes all families. The lower-cased back-off consumed the exact remaining budget perfectly.
- Model 2 ($\lambda=1.0$): Dropped sparse word-context features to fund high-impact signals (suffixes, shapes, clusters) under the 500-parameter limit.
- Decoding: memm_viterbi uses a top-50 state beam. Unseen words follow a strict fallback chain: tags seen with word $\rightarrow$ word shape $\rightarrow$ longest suffix $\rightarrow$ all tags.

4. Evaluation

- Model 1 scores 95.93% on the held-out test1.wtag.
- Model 2 achieves $93.0\% \pm 0.24$ in repeated 5-fold CV. The cluster feature consistently adds +0.3%. Extrapolating the learning curve (Fig. 1) yields 93.8% on the full set. Since OOV rates match perfectly, I predict $\approx 93.5\%$ on comp2.

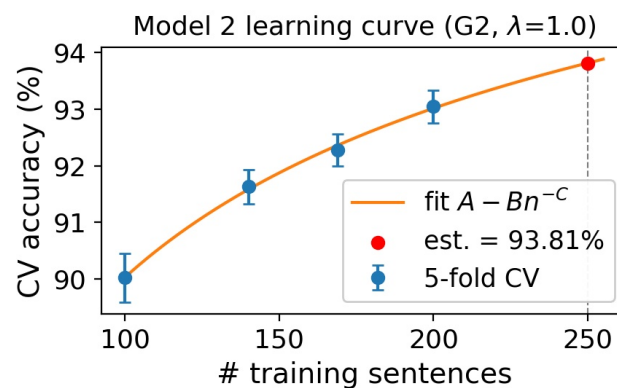


Figure 1. Model 2 CV accuracy vs. training size. Extrapolates to 93.8%.